



CS 546: Design of Concurrent Software

Course Code	CS 546
Course Name	Design of Concurrent Software
L-T-P-C	3-0-0-3
Prerequisites	programming & data structures; basics of operating systems, basics of networks.
Intended for	BTech, MTech, MSc, MS/PhD in the area.
Distribution	Discipline elective for BTech CSE and EE, free elective for others
Approval	10th Senate (Jun 2016)

Course Contents

- **Introduction:** Need for concurrent programs; the critical section problem; parallel and distributed architectures.
- **Programming models:** levels of parallelism; data distribution for arrays; shared variables and message-passing; processes and threads.
- **Performance evaluation:** metrics for parallel programs; design of experiments, measurement techniques, confidence levels.
- **Parallel programming:** the MPI message-passing model; point-to-point and collective communication modes; process groups; MPI and Pthreads; testing for correctness and for performance; debugging. Optionally: CUDA/OpenCL for GPU programming.
- **Concurrency in Java:** Java memory model; threads; RMI; locking; scalability; selected concurrency design patterns.
- **Big Data Processing:** the Map Reduce programming model; Map Reduce architecture and implementations; Map Reduce algorithm design; limitations of the Map Reduce model, extensions to solve these.

Text Books

1. T. Rauber & G. Rnger, **Parallel Programming for Multicore and Cluster Systems**, Springer, 2007.
2. B. Goetz et al., **Java Concurrency in Practice**, Pearson, 2006.
3. J. Lin & C. Dyer, **Data-Intensive Text Processing with Map Reduce**, Morgan & Claypool, 2010

References

1. E.D. Lazowska et al., **Quantitative System Performance**, Prentice-Hall, 1984.
2. K.S. Trivedi, **Probability and Statistics with Reliability, Queueing and Computer Science Applications**, Prentice-Hall, 1982.
3. Doug Lea, **Concurrent Programming in Java: Design Principles and Patterns**, 2nd Edition, Pearson, 2000.
4. M. Subramanian, **Network Management: Principles and Practice**, 2nd Edition, Pearson, 2009. (Chap 9.4)
5. D. Kirk and W. Hwu, **Programming Massively Parallel Processors: A Hands-on Approach**, 2nd Edition, Morgan Kaufmann, 2012.